

Yogita Bisht

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LinkedIn • GitHub • Portfolio

EDUCATION

Northeastern University, Massachusetts, USA

Sep 2024 – May 2026

Master of Science in Data Science, GPA: 3.6/4.0

Visvesvaraya Technological University, Karnataka, India

Aug 2017 – Aug 2021

Bachelor of Engineering in Computer Science and Engineering, CGPA: 8.5/10

WORK EXPERIENCE

Northeastern University

May 2025 – Aug 2025

Data Science Research Assistant, Skin Cancer Detection

Boston, MA

- Achieved **95% sensitivity** and **0.96 AUC-ROC** on melanoma detection by training and benchmarking four hybrid CNN-Vision Transformer architectures in PyTorch Lightning.
- Recovered minority-class performance on a severely imbalanced **400K+** image dataset by applying weighted sampling and targeted augmentation.

Mercedes-Benz R&D

Oct 2021 – Jul 2024

Engineer, After-Sales Diagnostics

Bangalore, India

- Reduced undetected diagnostic errors by **20%** by building **AWS Glue** ETL jobs that parsed and normalised **500GB+** of vehicle ECU fault logs.
- Resolved **30+** critical diagnostic failures affecting worldwide workshop technicians through collaboration across global engineering, product, and field operations teams.
- Raised automated test coverage from **70%** to **95%** by integrating regression test suites into **CI/CD** pipelines for agile diagnostic feature releases.
- Cut field issue resolution time by **25%** by building automated **SQL** pipelines and interactive dashboards for root-cause analysis.

Capgemini Engineering

Mar 2021 – Sep 2021

Data Analyst Intern

Bangalore, India

- Identified the top purchase drivers across **100K+** customer records by running **EDA** on **50+** features and applying **SHAP** for model interpretability.
- Cut manual data preparation by **4 hrs/week** by delivering **Power BI** dashboards that consolidated multi-source customer data for business stakeholders.

PROJECTS

Personalized Financial Recommendation System

- Built an end-to-end **Streamlit**-based recommendation engine for **26K+ financial products**, combining **K-Means** persona clustering with vector-based **cosine similarity** profile matching.
- Trained an **XGBoost** risk-tolerance classifier achieving a **0.94 F1-score** tracked via **MLflow**, and embedded real-time model predictions into a **RAG pipeline** to generate personalized explanations.

AskNEU – RAG-Based Conversational AI Assistant

- Developed a RAG-based chatbot using **LangChain** and **Hugging Face** to answer student questions, indexing **500+** web pages into **Pinecone Vector DB** and achieving **95%** retrieval accuracy.
- Automated ingestion with **Airflow** and deployed on **GCP Cloud Run** via **Docker** and **GitHub Actions**, with **LangSmith** for model monitoring and LLM observability.

TECHNICAL SKILLS

ML & GenAI | Python, Scikit-learn, PyTorch, LLMs, RAG, LangChain, Vector DB

MLOps & Cloud | Git/Github, Bash/Linux, FastAPI, Docker, Airflow, AWS, GCP

Data & Tools | Excel, SQL, Power BI, PySpark, Streamlit